



St Ignatius' College Riverview

Mathematics Assessment Task I Year 12

(Time allowed – 60 minutes)

- INSTRUCTIONS:**
- * Answer each question on a separate answer sheet.
 - * Write your name and your teacher's name on each answer sheet.
 - * Answer ALL questions.
 - * Approved calculators may be used.

QUESTION 1 : SEQUENCES & SERIES (20 marks)

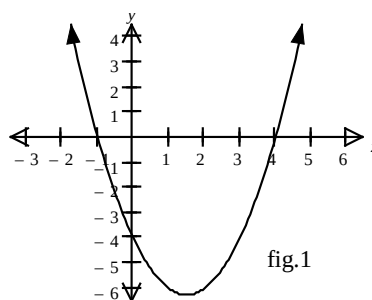
- a) The formula for the ***n*th term** of an arithmetic sequence is : $T_n = 2n + 4$
- i) What are the **4th** and **5th** terms? (2 marks)
- ii) What is the **common difference**? (1 mark)
- b) Find the values of x such that $\{3, x + 4, x + 10, \dots\}$ forms a geometric sequence. (2 marks)
- c) The 3rd term of an arithmetic progression is 16, and the 12th term is 79.
- i) Find the **first term** and **common difference**. (2 marks)
- ii) Find the **sum** of the first 25 terms. (2 marks)
- d) A super-ball drops from a height of 9 metres and bounces continually, each successive height being $\frac{2}{3}$ of the previous height.
- i) Show that the first distance travelled down and up is 15 metres. (1 mark)
- ii) When the ball finally comes to rest, through what distance will it have travelled in total? (2 marks)
- e) How many terms of the series $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ give a sum of $\frac{1023}{1024}$? (3 marks)

- f) The price of windows in a house is \$500 for the first window, then \$300 for each additional window.
- Find a formula for the cost of n windows. (1 marks)
 - How much will fifteen windows cost? (2 marks)
 - What is the maximum number of windows whose total cost is less than \$10,000? (2 marks)

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QUESTION 2 : THE QUADRATIC FUNCTION (20 marks)

- a) By observing the roots in fig. 1, write the equation of the parabola. (2 marks)

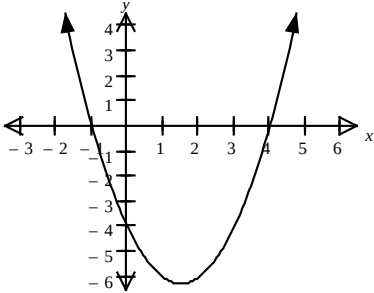


- b) Show that equation $2x^2 + x + 4 = 0$ has no real roots. (3 marks)
- c) Find all values of k for which the expression $kx^2 + 3kx + 6$ is positive definite. (3 marks)
- d) If α and β are the roots of the quadratic equation $x^2 - 3x - 6 = 0$, find the value of:
- $\alpha + \beta$ (1 mark)
 - $\alpha\beta$ (1 mark)
 - $\frac{1}{\alpha} + \frac{1}{\beta}$ (2 marks)
 - $\alpha^2 + \beta^2$ (2 marks)
- e) Solve: $x^2 + \frac{4}{x^2} = 5$ (3 marks)
- f) Find values of a , b and c if $x^2 - x \equiv a(x + 3)^2 + bx + c - 1$ (3 marks)

Solutions

a)	The formula for the <i>n</i>th term of an arithmetic sequence is : $T_n = 2n + 4$		Markers Comments
i)	What are the 4th and 5th terms? $T_4 = 2(4) + 4$ $T_4 = 12$ $T_5 = 2(5) + 4$ $T_5 = 14$	1 1	
ii)	What is the common difference ? $d = T_5 - T_4$ $d = 14 - 12$ $d = 2$	1	
b)	Find the value of x such that $\{3, x + 4, x + 10, \dots\}$ forms a geometric sequence. $\frac{x + 10}{x + 4} = \frac{x + 4}{3}$ $3(x + 10) = (x + 4)(x + 4)$ $3x + 30 = x^2 + 4x + 4x + 16$ $3x + 30 = x^2 + 8x + 16$ $x^2 + 5x - 14 = 0$ $(x + 7)(x - 2) = 0$ $x = -7 \text{ or } 2$	1 1	
c)	The 3 rd term of an arithmetic progression is 16, and the 12 th term is 79. i) Find the first term and common difference . $a + 2d = 16 \dots\dots\dots(1)$ $a + 11d = 79 \dots\dots\dots(2)$ $Eqn(2) - Eqn(1)$ $9d = 63$ $d = 7$ Sub $d = 7$ into $Eqn(1)$ $a + 2(7) = 16$ $a = 2$ ii) Find the sum of the first 25 terms. $S_n = \frac{n}{2}(2a + (n - 1)d)$ $S_{25} = \frac{25}{2}(2(2) + (25 - 1)7)$ $S_{25} = 2150$	1 1 1 1	
d)	A super-ball drops from a height of 9 metres and bounces continually, each successive height being $\frac{2}{3}$ of the previous height.		

i) Show that the first distance travelled down and up is 15 metres. Distance down once = 9 metres d) i)Continued. Distance up once = $9 \times \frac{2}{3}$ = 6 $\therefore$ Total distance = 9+6 = 15 metres ii) When the ball finally comes to rest, through what distance will it have travelled in total? $S_{\infty} = \frac{a}{1-r}$ Total distance = $2\left(\frac{6}{1-\frac{2}{3}}\right) + 9$ Total distance = 45 metres	1 1 1	
e) How many terms of the series $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ give a sum of $\frac{1023}{1024}$? $a = \frac{1}{2}, r = \frac{1}{2}$ $\frac{a(1-r^n)}{1-r} = \frac{1023}{1024}$ $\frac{\frac{1}{2}(1-\frac{1}{2}^n)}{1-\frac{1}{2}} = \frac{1023}{1024}$ $1-\frac{1}{2}^n = \frac{1023}{1024}$ $\frac{1}{2}^n = \frac{1}{1024}$ $(2^{-1})^n = 2^{-10}$ $2^{-n} = 2^{-10}$ $n = 10$	1 1 1	
f) The price of windows in a house is \$500 for the first window, then \$300 for each additional window. i) Find a formula for the cost of n windows. $T_n = 200 + 300n$ ii) How much will fifteen windows cost? $T_n = 200 + 300n$ where $n = 15$ $T_{15} = 200 + 300(15)$ $T_{15} = 4700$ $\therefore$ 15 widows will cost \$4700 iii) What is the maximum number of windows	1 1 1	

whose total cost is less than \$10,000? $10000 = 200 + 300n$ $9800 = 300n$ $n = 32.\bar{6}$ $\therefore 32 \text{ windows}$	1 1	
2a) By observing the roots in fig. 1, write the equation of the parabola. $\alpha = -1 \text{ and } \beta = 4$ $(x - \alpha)(x - \beta) = 0$ $\therefore (x + 1)(x - 4) = 0$ $\therefore x^2 - 4x + x - 4 = 0$ $\therefore x^2 - 3x - 4 = 0$	 1 1	
b) Show that equation $2x^2 + x + 4 = 0$ has no real roots. No real roots $\therefore \Delta < 0$ $b^2 - 4ac < 0$ $(1)^2 - 4(2)(4) < 0$ $1 - 32 < 0$ $-31 < 0$ $\therefore \Delta < 0$ $\therefore \text{No real roots}$	1 2	
c) Find all values of k for which the expression $kx^2 + 3kx + 6$ is positive definite. Positive Definite $\therefore a > 0$ and $\Delta < 0$ $b^2 - 4ac < 0$ $(3k)^2 - 4(k)(6) < 0$ $9k^2 - 24k < 0$ $3k(3k - 8) < 0$ $\therefore 0 < k < \frac{8}{3}$	1 1 1	
d) If α and β are the roots of the quadratic equation $x^2 - 3x - 6 = 0$, find the value of: i) $\alpha + \beta$ $\alpha + \beta = \frac{-b}{a}$ $\alpha + \beta = \frac{-(-3)}{1}$ $\alpha + \beta = 3$ ii) $\alpha\beta$	1	

$\alpha\beta = \frac{c}{a}$ $\alpha\beta = \frac{-6}{1}$ $\alpha\beta = -6$ iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta}{\alpha\beta} + \frac{\alpha}{\alpha\beta}$ $= \frac{\alpha + \beta}{\alpha\beta}$ $= \frac{3}{-6}$ $= -\frac{1}{2}$ iv) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= (3)^2 - 2(-6)$ $= 9 + 12$ $= 21$	1 1 1 1 1	
e) Solve: $x^2 + \frac{4}{x^2} = 5$ $x^4 + 4 = 5x^2$ $x^4 - 5x^2 + 4 = 0$ Let $m = x^2$ $\therefore m^2 - 5m + 4 = 0$ $\therefore (m - 4)(m + 1) = 0$ $\therefore m = 4 \text{ or } -1$ But $m = x^2$ $\therefore x^2 = 4 \quad \text{or} \quad x^2 = -1$ $\therefore x = \pm 2 \quad \text{or} \quad x = \pm\sqrt{-1} \text{ (invalid)}$ $\therefore x = 2 \text{ or } -2$	1 1 1	
f) Find values of a, b and c if $x^2 - x \equiv a(x + 3)^2 + bx + c - 1$ $a(x + 3)^2 + bx + c - 1 = a(x^2 + 6x + 9) + bx + c - 1$ $= ax^2 + 6ax + 9a + bx + c - 1$ $= ax^2 + (6a + b)x + 9a + c - 1$ For $x^2 - x \equiv a(x + 3)^2 + bx + c - 1$ $a = 1 \quad \text{.....(1)}$ $6a + b = -1 \quad \text{.....(2)}$ $9a + c - 1 = 0 \quad \text{.....(3)}$ Substitute (1) into (2)	1 1	

$6(1) + b = -1$ $b = -7$ Substitute (1) into (3) $9(1) + c - 1 = 0$ $c = -8$ $\therefore a = 1, b = -7, c = -8$	1	
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